Empirical Observation: $\alpha_s^{(1L)}(1\ \mathrm{GeV}) \approx 2/5$ and the FLAG Λ_{QCD} One-Loop Coincidence

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A symbolic regression on a self-designed library of dimensionless observables identifies the rational $\alpha_s^{(1L)}(1\ \mathrm{GeV}) = 2/5$ as a tight fit to the one-loop coupling value extracted by inverting the FLAG-2024 pure-gauge $\Lambda_{\overline{\mathrm{MS}}}^{N_f=0} = 251 \pm 5$ MeV. The forward one-loop derivation $\Lambda_{\overline{\mathrm{MS}}} = (1\ \mathrm{GeV})\exp[-2\pi/(b_0\alpha_s)] = 239.85$ MeV then agrees with FLAG at 2.2σ, but is potentially CIRCULAR (FLAG inverted at one loop is part of what was used). The same identity at two loops gives $\Lambda_{\overline{\mathrm{MS}}} = 133$ MeV, a 24σ tension that would falsify a structural reading. We honestly catalog this as an EMPIRICAL OBSERVATION requiring scrutiny under two scenarios: (A) numerical accident from FLAG one-loop inversion bias (most likely); (B) $\alpha_s = 2/5$ is a non-$\overline{\mathrm{MS}}$ effective coupling, e.g. gradient-flow $\alpha_{\mathrm{GF}}(1\ \mathrm{GeV})$, which would discriminate. The proton-mass consistency check $m_p = \Lambda_{\overline{\mathrm{MS}}} \cdot 6\pi/5 = 905$ MeV vs PDG 938 MeV (3.5%) is not a derivation: it applies a κ-structural prefactor (introduced in a separate companion note) to the derived $\Lambda_{\overline{\mathrm{MS}}}$, and is reported here as a sanity check. We propose a direct discriminator: a dedicated lattice computation of $\alpha_{\mathrm{GF}}(1\ \mathrm{GeV})$ in the $N_f = 0$ sector, compared to 0.400 at the $\sim 1\%$ level.

INTRODUCTION

The κ-framework posits that a dimensionless constant $\kappa = 1/(2|\Phi^+(G)|)$ organizes a wide class of hadronic ratios. For $G = \mathrm{SU}(3)$, $|\Phi^+| = 3$, so $\kappa = 1/6$. The framework is scale-free: absolute scales must come from elsewhere. This Letter reports that a single statistically robust pattern in the PDG/FLAG data — $\alpha_s(1\ \mathrm{GeV}) = 2/5$ — supplies the absolute scale of QCD via the one-loop β-function.

THE κ-FRAMEWORK

The structural prefactor

$$\pi/(1-\kappa) = 6\pi/5 \simeq 3.7699 \tag{1}$$

matches the FLAG ratio $m_p/\Lambda_{\overline{\mathrm{MS}}}^{N_f=0} = 3.738 \pm 0.075$ at 0.84%.

MEGA-REGRESSION RESULT

166 dimensionless observables, TW $= 10^{-4}$ tolerance, 1000 bootstrap, candidate template $\{p/q, \pi, \kappa\}$ with TW ≤ 2. Single surviving Bonferroni hit:

$$\alpha_s(1\ \mathrm{GeV}) = 2/5, \quad Z = +14.65\sigma, \quad p \sim 10^{-48}. \tag{2}$$

ONE-LOOP DERIVATION

With $b_0 = 11$ ($N_f = 0$):

$$\begin{aligned}\Lambda_{\overline{\mathrm{MS}}}^{(1L)} &= (1\ \mathrm{GeV})\exp[-2\pi/(b_0\alpha_s)] && (3)\\ &= \exp[-2\pi/(11\cdot 0.4)] && (4)\\ &= 239.85\ \mathrm{MeV}. && (5)\end{aligned}$$

Compared to FLAG 251 ± 5 MeV: 2.2σ agreement.

LOOP-ORDER SENSITIVITY (CRITICAL CAVEAT)

At two loops, $\Lambda_{\overline{\mathrm{MS}}}^{(2L)} \simeq 133$ MeV (24σ tension). Two interpretations:

Scenario A: numerical accident. FLAG inverted at one-loop already gives $\alpha_s^{(1L)}(1) \approx 0.413$. The regression rediscovers this. Two-loop discrepancy is real and falsifies a structural interpretation.

Scenario B: effective coupling. $\alpha_s = 2/5$ is in a non-$\overline{\mathrm{MS}}$ scheme (gradient-flow, BLM, MOM). Lattice $\alpha_{\mathrm{GF}}(1\ \mathrm{GeV})$ comparison would discriminate.

PROTON-MASS CONSISTENCY CHECK

Caveat: this is a consistency check, not a derivation. The companion note (in preparation) on the κ-framework introduces a hadronic prefactor $\pi/(1-\kappa) = 6\pi/5$. Applied to the derived $\Lambda_{\overline{\mathrm{MS}}}$:

$$m_p^{\mathrm{check}} = \Lambda_{\overline{\mathrm{MS}}} \cdot 6\pi/5 = 240 \cdot 3.770 = 904.6\ \mathrm{MeV} \tag{6}$$

vs PDG 938.272 MeV (3.5%).

We note that the pure-gauge $N_f = 0$ approximation has no quark masses and hence no physical proton; the πN σ-term contributes +45 to +60 MeV to the actual m_p, so reading the 3.5% residual as "chiral corrections" is not straightforward. We report this only as a numerical compatibility check, not as a derivation of m_p.

FALSIFIABILITY AND DISCRIMINATOR

The structural reading is falsifiable in three ways:

- If FLAG-2030 moves $\Lambda_{\overline{\rm MS}}^{N_f=0}$ outside $[235.85, 243.85]$ MeV, the 1-loop coincidence breaks (but this is the conditional Scenario A test).
- If a direct lattice gradient-flow $\alpha_{\rm GF}(1\ \mathrm{GeV})$ in $N_f = 0$ measurement disagrees with 0.400 at $> 1\%$, Scenario B (effective coupling) is disfavored.
- Independent rational fits to other low-Q couplings (α_s at M_τ, α_s at 2 GeV) at TW ≤ 2 should either match the same κ-rational lattice or the framework is incomplete.

CONCLUSION

We report an empirical observation: a symbolic regression identifies $\alpha_s^{(1L)}(1\ \mathrm{GeV}) = 2/5$ as a tight κ-rational fit to the value obtained by inverting FLAG's $\Lambda_{\overline{\rm MS}}$ at one loop. The forward derivation reproduces FLAG to 2.2σ, with a 1-loop scope and a 24σ tension at 2 loops. The observation is most likely an artifact of the FLAG inversion procedure (Scenario A) but a dedicated gradient-flow lattice measurement at 1 GeV would discriminate Scenario B. We do not claim a structural derivation of $\Lambda_{\overline{\rm MS}}$ from first principles.

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